Table 2. The percentage of novelty, clarity, feasibility, and relevance ratings ≥ 4 of proposed ideas. "#" means "the number of", and "bgs." is short for backgrounds. † means the proposed ideas are drawn from the paper's official code repository. ‡ means the running code is generated.

Domain	Methods	#Bgs. & #Ideas	LLM	Novelty	Clarity	Feasibility	Relevance	Helpful
NLP	GPT-4o	34 / 103	GPT-4o	16.50%	16.50%	18.45%	64.08%	38.83%
	AI Scientist†	3 / 30	GPT-4o	20.00%	(100%)‡	23.33%	-	46.67%
	SciPIP (Brainstorm)	34 / 123	GPT-4o	26.02%	34.15%	24.39%	62.60%	51.22%
	SciPIP (Retrieval)	34 / 167	GPT-4o	22.16%	31.74%	35.33%	**68.26%**	49.70%
	SciPIP	34 / 169	GPT-4o	**27.22%**	**36.69%**	**38.46%**	66.86%	**53.25%**
CV	GPT-4o	25 / 92	GPT-4o	13.04%	16.30%	14.13%	42.39%	47.83%
	SciPIP (Brainstorm)	25 / 97	GPT-4o-mini	20.62%	26.80%	24.74%	**47.42%**	72.16%
	SciPIP (Retrieval)	25 / 115	GPT-4o-mini	18.26%	18.26%	**26.96%**	43.48%	67.83%
	SciPIP	25 / 116	GPT-4o-mini	**22.41%**	**30.17%**	**26.72%**	**46.55%**	**74.14%**

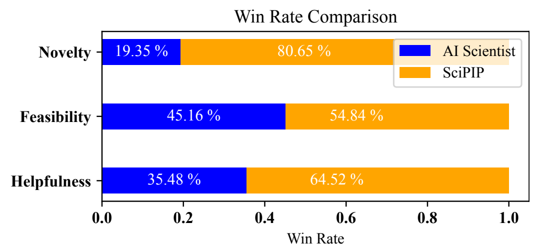

Figure 4. Comparison between SciPIP and AI Scientist. Both frameworks takes 3 backgrounds as input and generate about 30 ideas. GPT-4o-mini is used during generation.

4.2. Results and Analysis

Independent Ratings of Ideas. Through the construction of research backgrounds in both NLP and CV domains, we conduct a comprehensive comparison of proposal quality among SciPIP, GPT-4o, and AI Scientist. In this comparison, GPT-4o generates ideas directly through prompt engineering, utilizing the user's query as input. For AI Scientist, its ideas are either drawn from or generated using its official code repository, resulting in the use of different backgrounds compared to our approach.

All generated ideas were evaluated by invited researchers, and we calculated the percentage of ideas that received favorable ratings (≥ 4) from the evaluators. The results are presented in Table 2. As demonstrated, all three versions of SciPIP outperform GPT-4o and AI Scientist. Notably, SciPIP (Brainstorm) leverages internal knowledge similar to AI Scientist but produces ideas with significantly higher novelty. We hypothesize that this is because AI Scientist's input includes an existing paper and codebase, which may constrain its creative scope. The clarity of AI Scientist's ideas is consistently rated highly, as it generates executable code that provides clear explanations of the proposed ideas. Furthermore, the results reveal that brainstorming tends to generate more novel ideas compared to literature retrieval-based generation (25.77% vs. 22.14%), though the former often lacks sufficient feasibility (24.14% vs. 35.33%). SciPIP integrates the outputs of both brainstorming and retrieval-based generation, achieving an improved balance between novelty and feasibility while yielding improved results.

SciPIP also demonstrates superior performance in the CV domain, highlighting its generalizability across fields. However, its performance in CV is slightly inferior to that in NLP, particularly in terms of clarity. We attribute this discrepancy to the relatively weaker performance of GPT-4o-mini compared to GPT-4o. An intriguing observation is that 74.17% of CV ideas are rated as helpful, surpassing the corresponding percentage for NLP ideas (53.14%). This discrepancy may stem from inconsistent evaluation standards, as the evaluation tasks were distributed according to the volunteers' research fields, resulting in CV and NLP ideas being assessed by two distinct groups of researchers.

Win Rates between SciPIP vs. AI Scientist. We conducted a comparative analysis of the win rates between SciPIP and AI Scientist. AI Scientist is designed to generate novel ideas by leveraging existing papers, code bases, and experimental data as input, without requiring predefined challenges or problems. To ensure a fair comparison under identical conditions, we extracted the proposed ideas from AI Scientist and utilized an LLM to infer the underlying problems they aimed to address. These inferred backgrounds were then used as input for SciPIP, enabling a direct comparison of the ideas generated by both systems within the same contextual framework. The comparative win rates are illustrated in Figure 4. The results demonstrate that SciPIP consistently outperforms AI Scientist across all evaluation metrics. Notably, human researchers rated 80% of the ideas generated by SciPIP as more novel compared to those produced by AI Scientist. Additionally, SciPIP exhibits superior performance in terms of feasibility and helpfulness, further underscoring its effectiveness in generating high-quality research ideas.

Distribution of Ratings. The distribution of human ratings for NLP ideas generated by SciPIP is presented in Figure 5. The results demonstrate that ratings ≥ 3 consti-

Table 3. Demonstration of a SciPIP proposed idea. Only the concise version is provided due to space limitation, and the detailed version is in the Appendix B.

Background	1. The need to overcome the limitations of traditional RAG systems, which struggle with processing lengthy documents and filtering out irrelevant content, leading to inaccurate response generation. 2. The desire to eliminate the suboptimal chunking process, which disrupts semantic coherence and results in incomplete and incoherent retrieved information.
Proposed Idea	**Multi-Stage Contextual Thinning with Semantic Twin Projections** Proposed by integrating Dynamic Attention-Based Document Thinning and Semantic Twin Networks, introduce a multi-stage thinning process that refines document content leveraging semantic projections. Initially, a broad stroke thinning reduces document length by filtering with dual embeddings while maintaining high-level context. Subsequently, targeted semantic twin projections help identify and concentrate on key chunks according to their contextual relevance. This dual-phase smart thinning reduces the need for arbitrary chunking, preserving semantic connections and enhancing the efficiency and quality of generated outputs by guiding the generative model to produce well-informed, coherent responses.
Detailed Idea	Please refer to Appendix B for details.

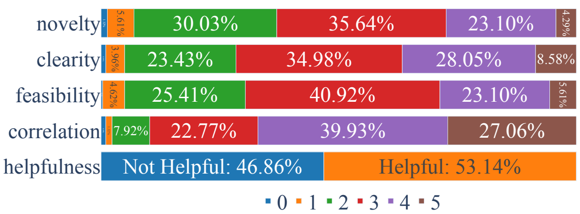

Figure 5. The distribution of human ratings of SciPIP proposed NLP ideas. GPT-4o is used during generation.

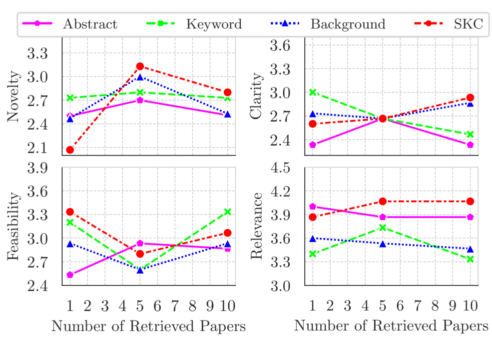

Figure 6. Ablation studies on different literature retrieval methods and the number retrieved papers. Y-axis means the average ratings over all ideas.

tute the majority across the novelty, clarity, and feasibility metrics. Given that the rating scale is divided into six segments, a score of 3 represents a relatively positive evaluation. Furthermore, over 66% of the ideas achieve relevance scores ≥ 4, indicating that SciPIP demonstrates strong capability in generating ideas that are well-aligned with the provided background. Additionally, approximately 53.14% of the ideas are rated as helpful, suggesting that more than half of the generated ideas are considered valuable by human evaluators.

Ablations on Retrieval Methods. We perform ablation studies to evaluate the impact of various literature retrieval methods and the number of retrieved papers on the overall performance. GPT-4o-mini is employed for idea generation in these experiments. We do not use larger models like GPT-4o because of the efficiency concern. The corresponding results are presented in Figure 6. The terms "Abstract" and "Background" refer to the encoding of the abstract section or our extracted background information as the retrieval vector, followed by vector-based retrieval. "Keyword" denotes the application of keyword-based retrieval exclusively. "SKC" represents our proposed multi-granularity retrieval method. As demonstrated in the results, SKC achieves the most balanced and superior performance across the evaluated metrics. No significant correlation is observed between the number of retrieved papers and the ratings. Therefore, to maintain a balance among the four metrics, we consistently retrieve 10 papers in subsequent experiments.

Case Study. Among the 169 ideas generated, as summarized in Table 2, several achieved high evaluation scores. One such idea, rated 4 for both novelty and feasibility, is presented in Table 3, while additional examples are included in the supplementary materials. The highlighted idea addresses a key limitation of existing retrieval-augmented generation (RAG) systems, which often rely on simplistic chunking methods. The proposed idea introduces a representation-based importance metric and semantic twin networks to selectively extract key information from lengthy documents. This approach is expected to enhance the performance of current RAG algorithms.

5. Conclusions and Limitations

In this paper, we propose a novel method, SciPIP, for generating scientific paper ideas and demonstrate its effectiveness in the domains of natural language processing and computer vision. Experimental results indicate that SciPIP leverages the capabilities of large language models (LLMs) to generate numerous innovative ideas. These ideas exhibit strong potential in terms of novelty, feasibility, clarity, and other critical dimensions. However, despite these promising results, the study has certain limitations. Specifically, the feasibility of the generated ideas was evaluated subjectively based on humans. Moreover, this study focuses on two specific fields including NLP and CV, and we intend to extend SciPIP to other fields and disciplines to assess and demonstrate its broader applicability and generalizability.

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